



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Experiment-1.3

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Subject Name: DAA

Subject Code: 23CSH-301

- 1. Aim:** Code to find frequency of elements in a given array in $O(n)$ time complexity.
- 2. Objective:** To find frequency of elements in a given array in $O(n)$ time complexity.
- 3. Input/Apparatus Used:** In this program, HashMap concept is used in order to get less complexity.

4. Procedure/Algorithm: Pseudocode:

Algorithm to find the frequency of each element of an array :

- Input the number of elements of an array.
- Input the array elements.
- Create another array to store the frequency of elements.
- Traverse the input array and update the count of the elements in the frequency array.
- Print the frequency array which displays the frequency of all the elements of the array.

Algorithm to find the frequency of each element of an array using hashing

- Input the number of elements of the array.
- Input the array elements.
- Create a hash table and update the element in one column and its frequency in the other column.
- Print the element along with its frequency.

Algorithm :

1. Create an extra space of size n (hm), use it as a HashMap.
2. Traverse the array from start to end.
3. For every element update $hm[array[i]-1]$, i.e. $hm[array[i]-1]++$

4. Run a loop from 0 to n and print hm[array[i]-1] along with the index i

Sample Input:

- 1). arr[] = {10, 20, 20, 10, 10, 20, 5, 20}
2). arr[] = {10, 20, 20}

Sample Output:

- 1). 10 3
 20 4
 5 1

2). 10 1
 20 2

5. Code:

```
#include <iostream>
#include <unordered_map>
using namespace std;

void findFrequency(int arr[], int n) {
    unordered_map<int, int> freq;
    for (int i = 0; i < n; i++) {
        freq[arr[i]]++;
    }
    cout << "Element    Frequency\n";
    for (auto it : freq) {
        cout << it.first << "    " << it.second << endl;
    }
}

int main() {
    int n;
    cin >> n;
    int arr[n];
    for (int i = 0; i < n; i++)
        cin >> arr[i];
    findFrequency(arr, n);
    return 0;
}
```

6. Output



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```
PS C:\Users\vaibh\OneDrive\Desktop\C tutorial 2> ./DAAExp3.exe
8
10
20
20
10
10
10
20
5
20
Element  Frequency
5      1
20     4
10     3
PS C:\Users\vaibh\OneDrive\Desktop\C tutorial 2> |
```